Seeing History Is Not Learning to Use It: An Exploratory Task-Inference Pilot

Abstract

An interactive policy must exploit evidence about an unknown task, not merely receive a long history. We investigate this distinction in the original DREAM Cooking environment using three recurrent behavior-cloning objectives with matched data, model size, initialization and optimizer updates. Over three optimizer seeds, observed-fact supervision achieves 56.80% success, compared with 24.21% for action-only supervision and 34.29% for current-state auxiliary supervision. Resetting task memory reduces factual-model success to 24.00%; a separately registered feedback-redaction diagnostic reduces it to 29.81%. Independent replay verifies all actions and outcomes, while a symbolic full-history reference remains at 100%. A second twelve-run factorial experiment shows that DAgger-style student-history action supervision reaches 81.33%, while factual supervision on student histories reaches 78.51%, rejecting its necessity at this budget. These are exploratory learning diagnostics, not a new auxiliary or distillation algorithm, robot result or conference-readiness claim.

1. Research Question and Prior Art

Tasks can share observations and control skills while differing in an unknown action–outcome mapping. Prior interaction may reveal the mapping even when it does not immediately finish the requested task. Our broader question is how experience should train the policy's task-inference process, in addition to its next action. A necessary first check is whether ordinary action supervision learns to exploit an informative history.

This question has substantial prior art. DREAM separates task-relevant exploration from exploitation (Liu et al., 2021). Task-belief supervision, including auxiliary prediction heads, was investigated by Humplik et al. (2019); VariBAD learns approximate inference and conditions behavior on uncertainty (Zintgraf et al., 2020). UNREAL already demonstrates representation benefits from auxiliary objectives (Jaderberg et al., 2017). Consequently, neither a factual prediction head nor the distinction between acquiring and using knowledge is claimed as an invention here. The pilot isolates a useful diagnostic before pursuing a narrower experience-update mechanism.

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2. Learning Object and Observable Boundary

Let τ index an unknown drawer mapping, and H_t denote the observed interaction prefix across attempts on that mapping. The recurrent policy state satisfies

$$m_t = U_\theta(m_{t-1}, x_t), \qquad \pi_\theta(a_t \mid m_t), \tag{1}$$

where x_t contains visible observation, revealed goal, previous actual action and reward, episode boundary, phase and local time. The mapping ID, hidden assignment, factual labels, teacher and symbolic memory are excluded from inference. State is retained across attempts on one task and cleared at the next task. An actual terminal observation and reward are ingested before each reset as a feedback-only token; no extra environment action is spent.

All arms use a 96-unit embedding and 96-unit causal GRU with identical action and auxiliary head sizes (63,576 parameters). The common training objective is

$$\mathcal{L}_{\text{action}} + 0.2\mathcal{L}_{\text{aux}}, \tag{2}$$

with the auxiliary term zero for action-only training. The current-state control predicts inventory and the first two visible bowl slots. The factual target reconstructs the latest observed food in each of three drawers, leaving an unobserved drawer unknown. Targets are generated exclusively from actual pickup observations at or before the input prefix. They are never fed to the policy.

3. Protocol and Comparability

We pin the official DREAM implementation to revision `d52204c` (full identity in the plan), preserving transition, reward, observation and the 20-step horizon. Our explicitly registered 32/8/24 mapping split differs from the original paper's setup and reuses our earlier diagnostic panel. This is not a DREAM performance reproduction or an untouched confirmatory test.

Table 1. Five-attempt mean success (%). Three optimizer seeds are reported completely. Reset uses the same checkpoint with task memory cleared between episodes.

Objective	Mean	Seed range	Reset
Action-only	24.21	20.00–27.84	18.67
Current-state aux.	34.29	31.84–36.48	18.13
Observed-fact aux.	56.80	51.68–60.80	24.00
Symbolic reference	100.00	100.00–100.00	–

The common training set has 1,062 histories over 32 mappings and 177 valid goals: three behavior seeds, each with deterministic and fixed 10% action-perturbation collection. The teacher acts only on observed facts. This provides limited recovery-state coverage, not student-on-policy data. For each optimizer seed (7, 19, 41), the three arms share initial weights and minibatch indices. Each receives 800 Adam updates at learning rate 0.003, batch size 24 sequences and gradient clipping 1. The final checkpoint is fixed; development does not select it. Nine runs spend 7,200 optimizer updates.

Each evaluated task includes up to 20 exploration actions followed by five executions of the same goal. The learned policy controls from the initial step. The 24 evaluation mappings provide 125 valid mapping–goal cases; each arm contributes 1,875 execution episodes across three checkpoints. Repeated episodes are nested under mapping and optimizer seed, not independent training replicates. A failed execution receives cost 20; actual action counts are also recorded.

4. Results and Separating Diagnostics

Factual supervision exceeds action-only by 32.59 percentage points and the current-state control by 22.51 points, with positive paired differences in all three seeds. Memory reset causes a 32.80-point drop. These meet the registered pilot predictions (five-point differences, positive direction per seed and a five-point memory effect). Three seeds support descriptive comparisons, not a general superiority test. High teacher-forced training accuracy is not used as closed-loop efficacy evidence.

Factual success rises only from 54.93% on the first execution to 57.60% on the fifth. The first execution already follows 20 interaction steps and is not zero-shot. The strongest evidence concerns using acquired information, rather than progressively better experimentation or recursive self-improvement.

Shared history and feedback use. After reading the primary results, we register a second experiment using the nine frozen checkpoints. Each receives a common real 20-step exploration prefix. The paired intervention replaces only prefix inventory observations with EMPTY at learner input; execution observations remain unchanged. Raw logs retain original observations. This supplied-prefix diagnostic is distinct from primary teacher-free exploration and uses no gradients.

Independent comparison finds that all 1,125 autonomous prefixes already equal the supplied common prefix. Thus, differences in exploration information do not explain the primary comparison on these cases. Redaction collapses 125 prefix input sequences to one. The complete results are: The factual model uses historical feedback; merely possessing a recurrent history is insufficient for the action-only baseline. Redaction slightly improves action-only success and changes the input distribution. We therefore do not interpret the intervention as universal benefit from memory or as proof of a new causal-credit theory.

Table 2. Shared-prefix feedback intervention (%). Registration occurred after the primary pilot.

Model	Real	Redacted	Difference
Action-only	24.21	26.99	−2.77
Current-state aux.	34.29	32.91	+1.39
Observed-fact aux.	56.80	29.81	+26.99

5. Student-History Coverage as a Separating Test

A post-primary trace audit found that the recorded visible history contained the goal ingredients at every execution start, and every failed episode eventually corrupted the irreversible bowl. The factual decoder was correct for all three drawer slots on 24.72% of execution decisions. Even conditional on complete factual correctness, the action agreed with one of the six symbolic reference tie orders on only 64.38% of decisions. These are trace associations, not an identification of failure causes.

We therefore register a second 2×2 experiment crossing collection (offline perturbed teacher versus current student) with supervision (actions versus actions plus facts). All four arms start from the same action-only parent weights for each seed. Two rounds each add 177 training-only histories and 300 updates, maintaining Adam across rounds after its common restart. Student collection uses $\beta = 0$: every executed action comes from the current policy; the observable-memory teacher only labels the visited prefix. This is a DAgger-style comparator. GMPS already extends dataset aggregation to meta-learning (Mendonca et al., 2019); we do not claim to reproduce GMPS's gradient-based adaptation algorithm.

One common label-validity rule is registered before the new run: after an observed bowl contradicts the goal irreversibly,

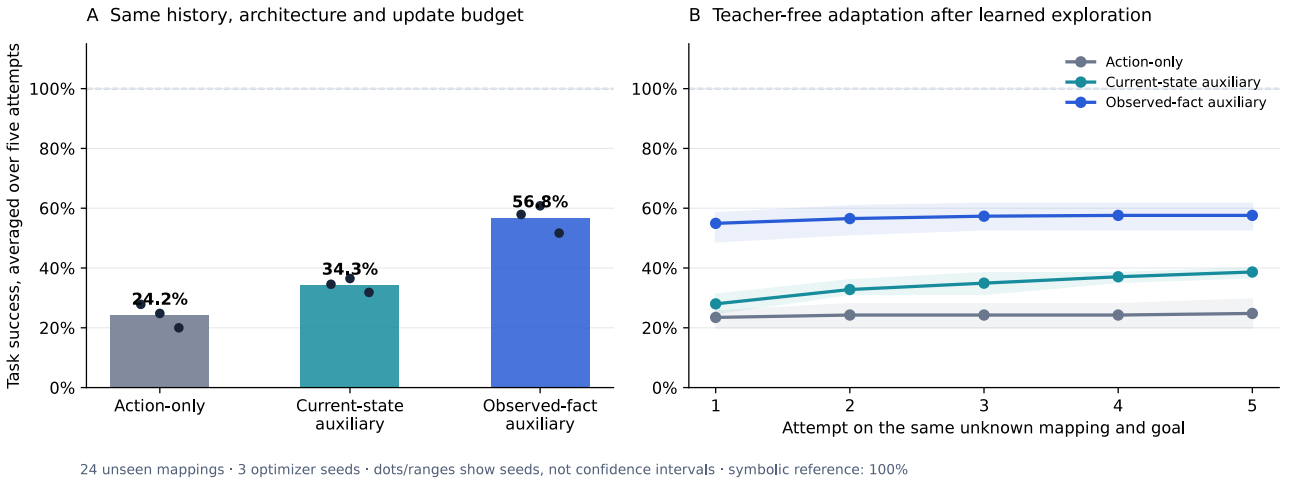


Figure 1. Matched learning budgets and complete adaptation curves. Dots and shaded ranges show the three optimizer seeds, not confidence intervals. Exploration is autonomous; the symbolic reference is a diagnostic ceiling. These results isolate auxiliary-learning behavior, not DREAM, AD or VariBAD performance.

Table 3. Student-history coverage versus factual supervision. Mean and all paired optimizer seeds, success in %. Each starts from the same parent and gets 600 additional updates.

Collection / objective	Mean	Seeds 7 / 19 / 41
Offline / actions	30.03	27.20 / 33.60 / 29.28
Offline / facts	60.53	59.36 / 65.60 / 56.64
Student / actions	81.33	66.72 / 94.56 / 82.72
Student / facts	78.51	81.92 / 82.56 / 71.04

action loss abstains while actual inputs and factual supervision remain. It is applied to all four arms' old and new histories. No preceding frames, task parameters or parent outcomes change. The panel does not isolate this guard's effect. Loss separation for failed behavior already has precedent in FACT (Peng et al., 2026).

Student-history action supervision exceeds offline action supervision by 51.31 percentage points, with positive paired differences in all seeds. Fact supervision adds 30.51 points offline but -2.83 points under student collection: $+15.20$, -12.00, and -11.68. The registered prediction of a five-point factual benefit persisting after student coverage fails. The descriptive interaction is -33.33 points. Ordinary distribution coverage and error correction remain the simpler explanation; the factual head is not established as a necessary missing mechanism.

New collection costs 373,297 physical actions and 25,488 resets; the twelve runs add 7,200 updates. Independent verification reconstructs all 4,248 histories, 202,711 student collector decisions from preserved intermediate checkpoints, and 18,000 final evaluation records with 292,359 actions. Raw observation differences are zero. There are 315,911 valid new correction labels and 57,386 ignored corrupted-state label sites. Candidate label sites are not measurements of external human or API queries. Different history lengths and label counts preclude an equal-token or budget-advantage claim. The evaluated panel and seeds are reused, so this remains an exploratory test.

6. Integrity, Costs and Failure History

The first registered learner run omitted terminal feedback from the GRU despite logging it. This could let the next episode's auxiliary target refer to unseen evidence. We interrupted that run before evaluation, retained its code and plan, and registered a correction adding actual feedback-only tokens. Task, split, horizon, model, loss weight, seeds and update budget did not change. No scientific metric uses that run. Its overhead is 79,904 source actions and 2,700–2,799 optimizer updates.

The valid source spends 79,904 physical actions and 86,276 input tokens, including 6,372 feedback-only tokens. Independent verification covers all 19,098 primary records, 322,100 evaluation actions and 295,280 neural decisions recomputed from checkpoints. The follow-up independently verifies 11,250 records and 185,786 policy decisions, plus 2,500 shared-prefix collection actions. Raw observation differences are zero; outcomes, rewards, identities and hash chains are reconstructed. This is reset-and-action replay, not snapshot restoration or a training reproducibility result. Fifteen original boundary tests and five new coverage and label-validity tests pass. No GPU or shared driver configuration is changed; training runs on CPU.

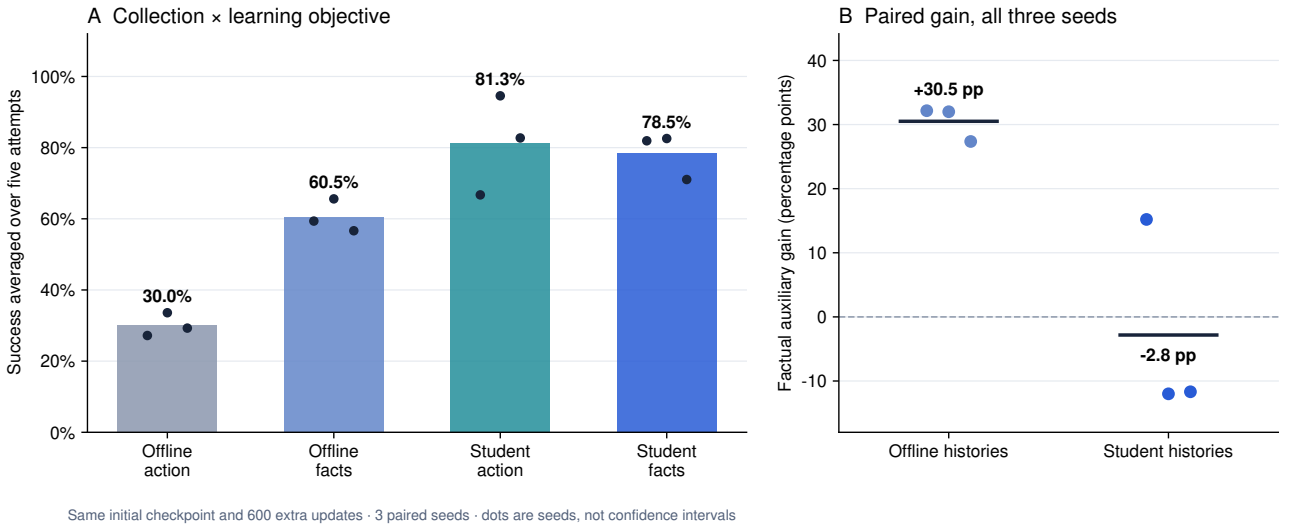


Figure 2. The previous auxiliary advantage does not survive student-history coverage in this fixed diagnostic. Every seed is shown; points are not confidence intervals. Collection actions and label counts differ, despite matching initial checkpoints, sequence counts, minibatch indices and updates. The plotted interaction is a system comparison, because the training objective also changes later collected histories.

7. What Remains Unestablished

The second experiment retains ordinary student-distribution coverage over the claim that a factual auxiliary head is essential. Joint belief-imitation learning is also prior art (Gangwani et al., 2020). Neither mechanism is established as a distinctive contribution. The symbolic reference remains at ceiling and only a finite mapping family is tested. The first three-objective experiment uses offline teacher histories; the second tests student-history aggregation. Student-history correction has now been tested as a baseline; no original OPD algorithm, learned exploration advantage, execution-failure discrimination, future-learning-credit objective, robot transfer or recursive acceleration is established.

A meaningful next comparison must include trained task-inference and full-history methods, controlled student-history labeling, and a strong observable-belief reference. A narrower candidate concerns whether an unsuccessful action supports changing a task hypothesis when execution itself is uncertain. That would require a separately registered environment extension, matched visible feedback and explicit controls for ordinary Bayesian inference; it cannot be inferred from this deterministic Cooking pilot. Global-task/local-state factorization already exists (Akuzawa et al., 2021), and belief-conditioned tactile probing has a contemporary robotics neighbor (Lee et al., 2026). Any candidate update target or future-adaptation selector must exceed these explanations, rather than merely rename them. Historical stopped Lift/Planar experiments remain separate.

Availability and status. Code, immutable plans, all-seed summaries, costs, verification and vector figures are at `https://asimfish.github.io/seec-research/learning.html`. Upstream DREAM code is not redistributed. This document is a working note in the repository's existing ICML 2026 format, not a completed method paper or submission.

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